

Delivery Package Index — t4_r1_noforce

Case: t4_r1_noforce (extreme-cold equipment battery, virtual battery factory)

Scheme: VBF-T4R1NOFORCE-<DOC-CODE>-<SEQ-NO>

Generation date: 2026-08-25

Prepared/Reviewed/Approved: ____ / ____ / ____

Delivery Package Index

- Case name: t4_r1_noforce (extreme-cold equipment battery, virtual battery factory)
- Numbering scheme: VBF-<CASE-ID-UPPERCASE>-<DOC-CODE>-<SEQ-NO> (case id uppercase, non-alphanumeric removed: t4_r1_noforce -> T4R1NOFORCE; SEQ-NO two-digit per doc code from 01)
- Generation date: 2026-08-25
- Prepared: ____ Reviewed: ____ Approved: ____

Document code reference (protocol-fixed)

Document code	Meaning	Corresponding file
DS	Specification	design_spec.md
BOM	Bill of Materials	bom.xlsx
DSH	Datasheet	datasheet.md
CALC	Calculation sheet	calc.xlsx
DVPR	Design verification report	dvpr.md
DFMEA	Failure analysis	dfmea.md
IDX	Delivery index	delivery_index.md
CAD	Structure model	not requested this case — skipped

File list

File name	Number	Format	Source description
design_spec.md	VBF-T4R1NOFORCE-DS-01	md	generated per deliverable-design-spec; values from Chen2020 set + params_r6_v10a_h45.json + round-6 tool outputs
design_spec.pdf	VBF-T4R1NOFORCE-DS-01	pdf	PDF release of design_spec.md (reportlab)
report.html	VBF-T4R1NOFORCE-DS-02	html	bda render of log.jsonl (ancillary to DS)
bom.xlsx	VBF-T4R1NOFORCE-BOM-01	xlsx	generated per deliverable-bom (openpyxl); layer masses from r6_final_energy_dfn.json, additive/binder literature defaults annotated
bom.pdf	VBF-T4R1NOFORCE-BOM-01	pdf	PDF release of bom.xlsx (reportlab)
datasheet.md	VBF-T4R1NOFORCE-DSH-01	md	generated per deliverable-datasheet
datasheet.pdf	VBF-T4R1NOFORCE-DSH-01	pdf	PDF release of datasheet.md (reportlab)
calc.xlsx	VBF-T4R1NOFORCE-CALC-01	xlsx	generated per deliverable-calc-sheet (openpyxl): input parameters -> capacity/energy -> energy density -> N/P and mass -> process parameters
calc.pdf	VBF-T4R1NOFORCE-CALC-01	pdf	PDF release of calc.xlsx (reportlab)
dvpr.md	VBF-T4R1NOFORCE-DVPR-01	md	generated per deliverable-dvpr (virtual test version)
dvpr.pdf	VBF-T4R1NOFORCE-DVPR-01	pdf	PDF release of dvpr.md (reportlab)
dfmea.md	VBF-T4R1NOFORCE-DFMEA-01	md	generated per deliverable-dfmea (qualitative version)
dfmea.pdf	VBF-T4R1NOFORCE-DFMEA-01	pdf	PDF release of dfmea.md (reportlab)
delivery_index.md	VBF-T4R1NOFORCE-IDX-01	md	this file
delivery_index.pdf	VBF-T4R1NOFORCE-IDX-01	pdf	PDF release of this file (reportlab, blueprint style)

